# E-COMMERCE SALES AND CUSTOMER ANALYSIS

**Presented by : Aleti Ravalika**

# Sample Data

| | Order Id | Date | Region | Salesperson | Product | Quantity Sold | Sales Amount | Discount (%) | Profit Margin (%) | Total Sales Amount |
|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | 2024-01-01 | East | Arun Kumar | Product X | 150.0 | 3000.0 | 10.0 | 20.0 | 450000.0 |
| **1** | 1002 | 2024-01-02 | West | Priya Sharma | Product Y | 200.0 | 4000.0 | 15.0 | 25.0 | 800000.0 |
| **2** | 1003 | 2024-01-03 | North | Rajesh Patel | Product X | 120.0 | 2400.0 | 5.0 | 18.0 | 288000.0 |
| **3** | 1004 | 2024-01-04 | South | Meera Reddy | Product Z | 250.0 | 5000.0 | 20.0 | 30.0 | 1250000.0 |
| **4** | 1005 | 2024-01-05 | East | Arun Kumar | Product Y | 100.0 | 2000.0 | 10.0 | 22.0 | 200000.0 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **85** | 1086 | 2024-03-27 | South | Meera Reddy | Product X | 130.0 | 2600.0 | 12.0 | 23.0 | 338000.0 |
| **86** | 1087 | 2024-03-28 | East | Arun Kumar | Product Y | 150.0 | 3000.0 | 13.0 | 24.0 | 450000.0 |
| **87** | 1088 | 2024-03-29 | West | Priya Sharma | Product Z | 200.0 | 4000.0 | 11.0 | 21.0 | 800000.0 |
| **88** | 1089 | 2024-03-30 | North | Rajesh Patel | Product X | 160.0 | 3200.0 | 8.0 | 22.0 | 512000.0 |
| **89** | 1090 | 2024-03-31 | South | Meera Reddy | Product Y | 180.0 | 3600.0 | 17.0 | 28.0 | 648000.0 |

# Displaying the First 5 Rows

```
df.head()
```

| | Order Id | Date | Region | Salesperson | Product | Quantity Sold | Sales Amount | Discount (%) | Profit Margin (%) | Total Sales Amount |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1001 | 2024-01-01 | East | Arun Kumar | Product X | 150.0 | 3000.0 | 10.0 | 20.0 | 450000.0 |
| 1 | 1002 | 2024-01-02 | West | Priya Sharma | Product Y | 200.0 | 4000.0 | 15.0 | 25.0 | 800000.0 |
| 2 | 1003 | 2024-01-03 | North | Rajesh Patel | Product X | 120.0 | 2400.0 | 5.0 | 18.0 | 288000.0 |
| 3 | 1004 | 2024-01-04 | South | Meera Reddy | Product Z | 250.0 | 5000.0 | 20.0 | 30.0 | 1250000.0 |
| 4 | 1005 | 2024-01-05 | East | Arun Kumar | Product Y | 100.0 | 2000.0 | 10.0 | 22.0 | 200000.0 |

# Displaying the Last 5 Rows

```
df.tail()
```

| | Order Id | Date | Region | Salesperson | Product | Quantity Sold | Sales Amount | Discount (%) | Profit Margin (%) | Total Sales Amount |
|---|---|---|---|---|---|---|---|---|---|---|
| **85** | 1086 | 2024-03-27 | South | Meera Reddy | Product X | 130.0 | 2600.0 | 12.0 | 23.0 | 338000.0 |
| **86** | 1087 | 2024-03-28 | East | Arun Kumar | Product Y | 150.0 | 3000.0 | 13.0 | 24.0 | 450000.0 |
| **87** | 1088 | 2024-03-29 | West | Priya Sharma | Product Z | 200.0 | 4000.0 | 11.0 | 21.0 | 800000.0 |
| **88** | 1089 | 2024-03-30 | North | Rajesh Patel | Product X | 160.0 | 3200.0 | 8.0 | 22.0 | 512000.0 |
| **89** | 1090 | 2024-03-31 | South | Meera Reddy | Product Y | 180.0 | 3600.0 | 17.0 | 28.0 | 648000.0 |

# Counting Missing Values in Each Column

```
df.isnull().sum()
```

```
Order Id                0
Date                    0
Region                  0
Salesperson             0
Product                 0
Quantity Sold           2
Sales Amount            2
Discount (%)            1
Profit Margin (%)       1
Total Sales Amount      6
dtype: int64
```

# Replacing Null Values with Mean

```
df=df.fillna(df.mean(numeric_only=True))
```

| | Order Id | Date | Region | Salesperson | Product | Quantity Sold | Sales Amount | Discount (%) | Profit Margin (%) | Total Sales Amount |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1001 | 2024-01-01 | East | Arun Kumar | Product X | 150.0 | 3000.0 | 10.0 | 20.0 | 450000.0 |
| 1 | 1002 | 2024-01-02 | West | Priya Sharma | Product Y | 200.0 | 4000.0 | 15.0 | 25.0 | 800000.0 |
| 2 | 1003 | 2024-01-03 | North | Rajesh Patel | Product X | 120.0 | 2400.0 | 5.0 | 18.0 | 288000.0 |
| 3 | 1004 | 2024-01-04 | South | Meera Reddy | Product Z | 250.0 | 5000.0 | 20.0 | 30.0 | 1250000.0 |
| 4 | 1005 | 2024-01-05 | East | Arun Kumar | Product Y | 100.0 | 2000.0 | 10.0 | 22.0 | 200000.0 |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 85 | 1086 | 2024-03-27 | South | Meera Reddy | Product X | 130.0 | 2600.0 | 12.0 | 23.0 | 338000.0 |
| 86 | 1087 | 2024-03-28 | East | Arun Kumar | Product Y | 150.0 | 3000.0 | 13.0 | 24.0 | 450000.0 |
| 87 | 1088 | 2024-03-29 | West | Priya Sharma | Product Z | 200.0 | 4000.0 | 11.0 | 21.0 | 800000.0 |
| 88 | 1089 | 2024-03-30 | North | Rajesh Patel | Product X | 160.0 | 3200.0 | 8.0 | 22.0 | 512000.0 |
| 89 | 1090 | 2024-03-31 | South | Meera Reddy | Product Y | 180.0 | 3600.0 | 17.0 | 28.0 | 648000.0 |

90 rows × 10 columns

# Verifying Missing Values After Imputation

```
df.isnull().sum()
```

```
Order Id                0
Date                    0
Region                  0
Salesperson             0
Product                 0
Quantity Sold           0
Sales Amount            0
Discount (%)            0
Profit Margin (%)       0
Total Sales Amount      0
dtype: int64
```

# Region-wise Cumulative Sales

```
df = df.sort_values(["Region", "Date"])

df["Cumulative Sales"] = (
    df.groupby("Region")["Sales Amount"]
      .cumsum()
)

df
```

| | Order Id | Date | Region | Salesperson | Product | Quantity Sold | Sales Amount | Discount (%) | Profit Margin (%) | Total Sales Amount | Cumulative Sale |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1001 | 2024-01-01 | East | Arun Kumar | Product X | 150.000000 | 3000.0 | 10.0 | 20.000000 | 450000.00000 | 3000.00000 |
| 4 | 1005 | 2024-01-05 | East | Arun Kumar | Product Y | 100.000000 | 2000.0 | 10.0 | 22.000000 | 200000.00000 | 5000.00000 |
| 8 | 1009 | 2024-01-09 | East | Arun Kumar | Product Z | 170.000000 | 3400.0 | 18.0 | 23.000000 | 578000.00000 | 8400.00000 |
| 12 | 1013 | 2024-01-13 | East | Arun Kumar | Product Y | 140.000000 | 2800.0 | 11.0 | 24.000000 | 392000.00000 | 11200.00000 |

# Total Sales Amount by Salesperson

```
df.groupby("Salesperson")["Sales Amount"].sum().reset_index()
```

| | Salesperson | Sales Amount |
|---|---|---|
| 0 | Arun Kumar | 71400.000000 |
| 1 | Meera Reddy | 84000.000000 |
| 2 | Priya Sharma | 74945.454545 |
| 3 | Rajesh Patel | 73200.000000 |

# Top-Selling Product by Region

```
df4 = (
    df.groupby(["Region", "Product"])["Quantity Sold"]
      .sum()
      .reset_index()
)

df4.loc[
    df4.groupby("Region")["Quantity Sold"].idxmax()
]
```

| | Region | Product | Quantity Sold |
|---|---|---|---|
| **2** | East | Product Z | 1428.068182 |
| **3** | North | Product X | 1280.000000 |
| **7** | South | Product Y | 1720.000000 |
| **11** | West | Product Z | 1508.068182 |

# Calculating Total Sales Using NumPy

```
sales = df["Sales Amount"].to_numpy()

print(np.sum(sales))
```

```
303545.45454545453
```

# Counting Region Occurrences Using NumPy

```
regions = df["Region"].to_numpy()

unique, counts = np.unique(regions, return_counts=True)

index = np.argmax(counts)

print(unique[index])
print(counts[index])
```

```
East
23
```

# Unique Region Analysis Using NumPy

```
regions = df["Region"].to_numpy()

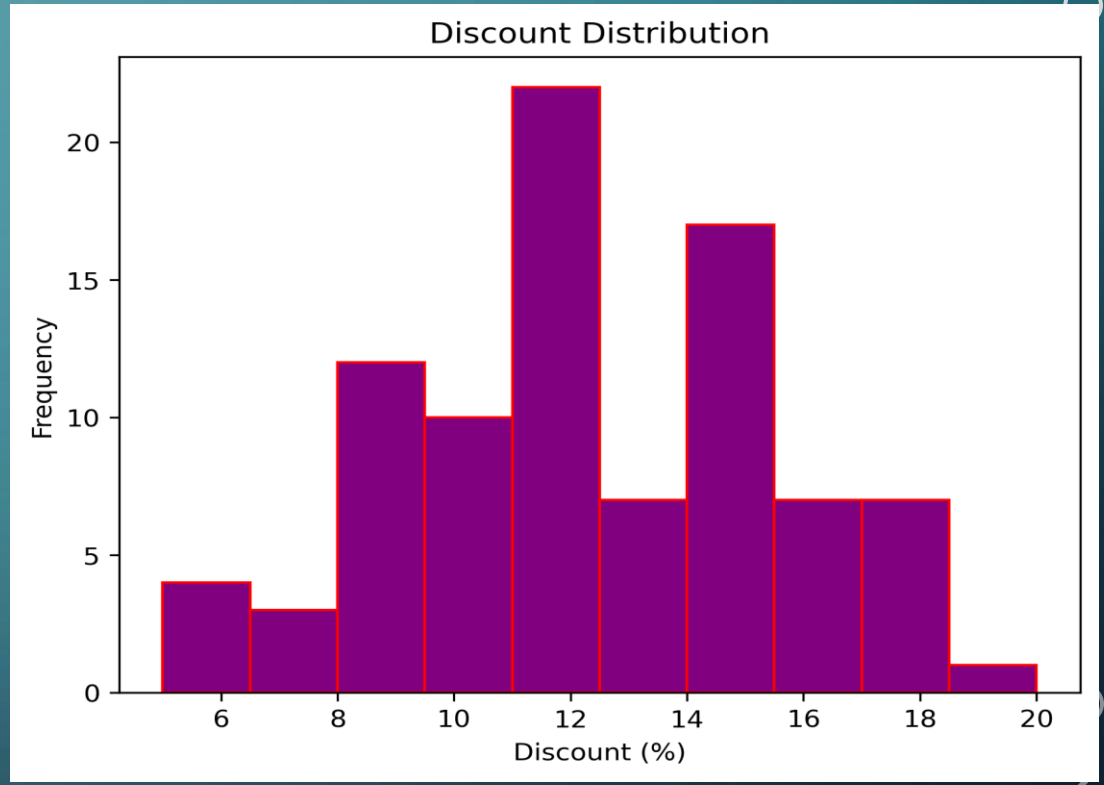
print(np.unique(regions))
print(len(np.unique(regions)))
```

```
['East' 'North' 'South' 'West']
4
```

# Discount Distribution Analysis

```
plt.hist(df["Discount (%)"], bins=10,color="purple",edgecolor="red")


plt.title("Discount Distribution")
plt.xlabel("Discount (%)")
plt.ylabel("Frequency")
plt.savefig("discount.png", dpi=300, bbox_inches="tight")
plt.show()
```



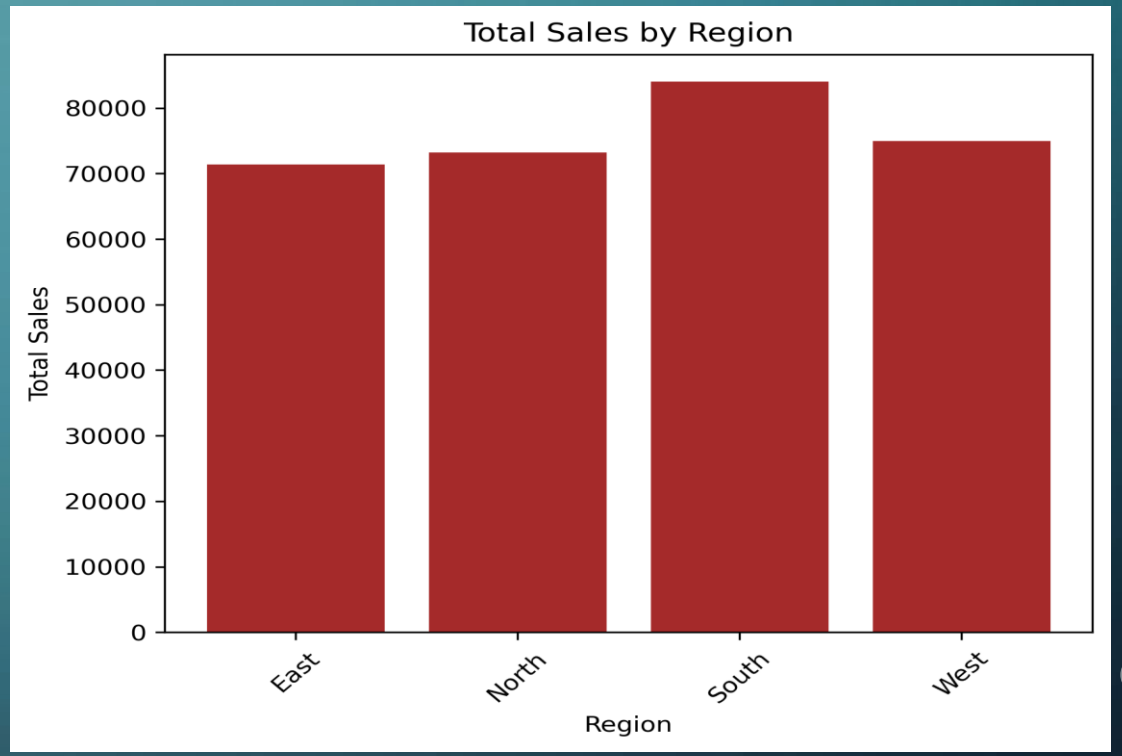
# Total Sales Analysis by Region

```
region_sales = df.groupby("Region")["Sales Amount"].sum()

plt.bar(region_sales.index, region_sales.values,color="brown")

plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")

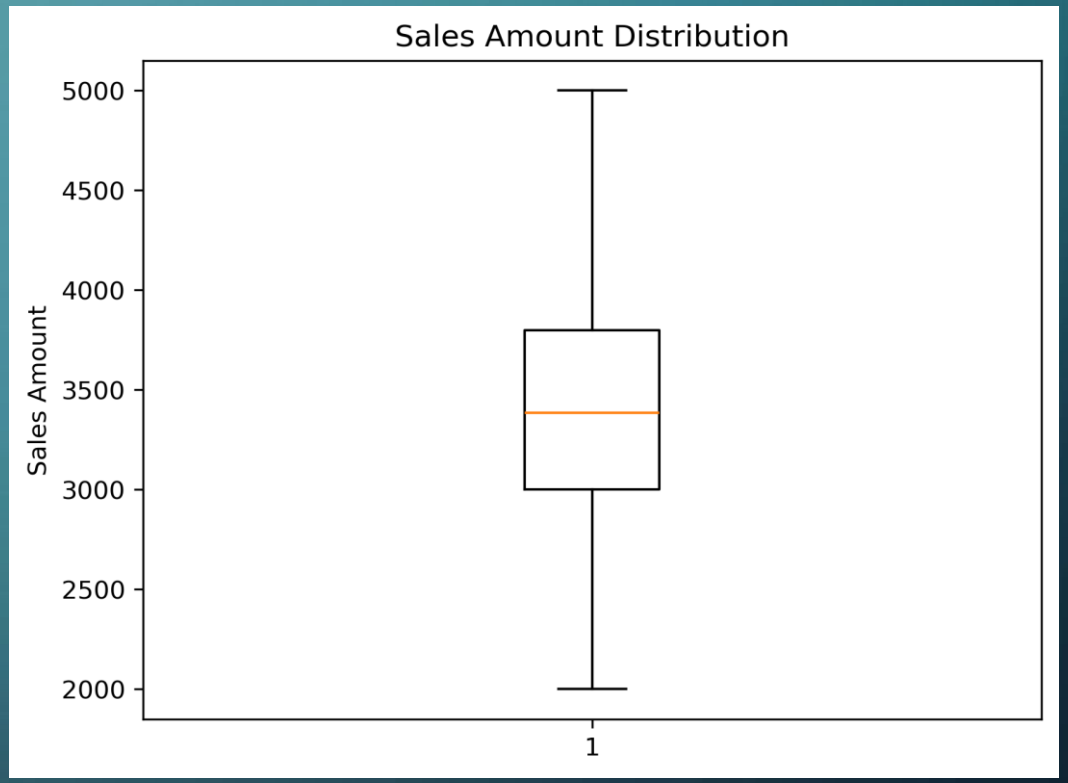
plt.xticks(rotation=45)
plt.savefig("sales_amount.png", dpi=300, bbox_inches="tight")
plt.show()
```



# Sales Amount Distribution Analysis

```
plt.boxplot(df["Sales Amount"])


plt.title("Sales Amount Distribution")
plt.ylabel("Sales Amount")
plt.savefig("sales.png", dpi=300, bbox_inches="tight")
plt.show()
```

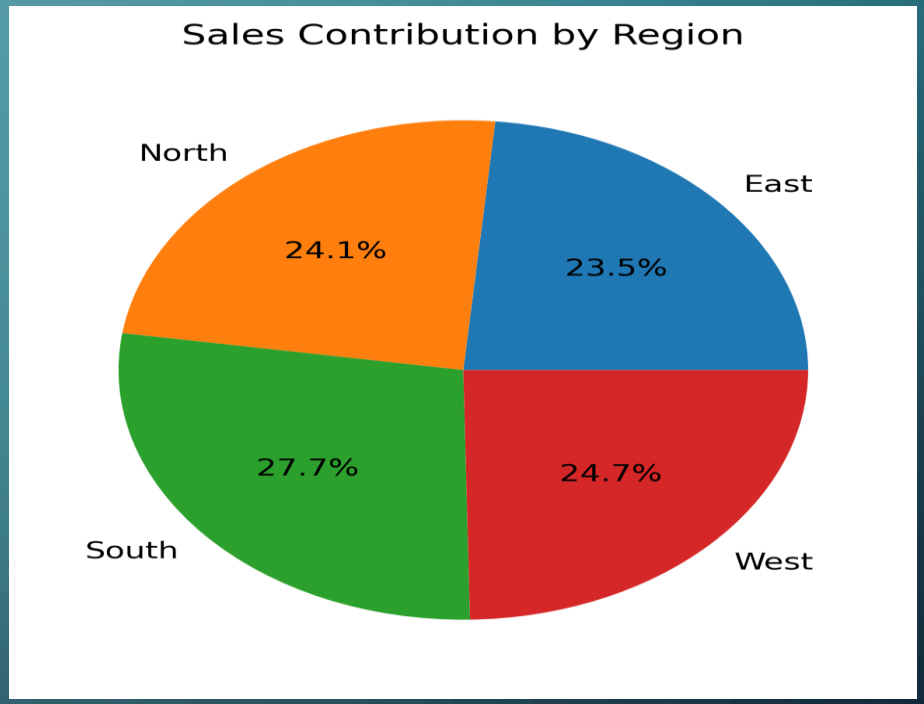


# Region-wise Sales Contribution

```
region_sales = df.groupby("Region")["Sales Amount"].sum()

plt.pie(
    region_sales,
    labels=region_sales.index,
    autopct="%1.1f%%"
)

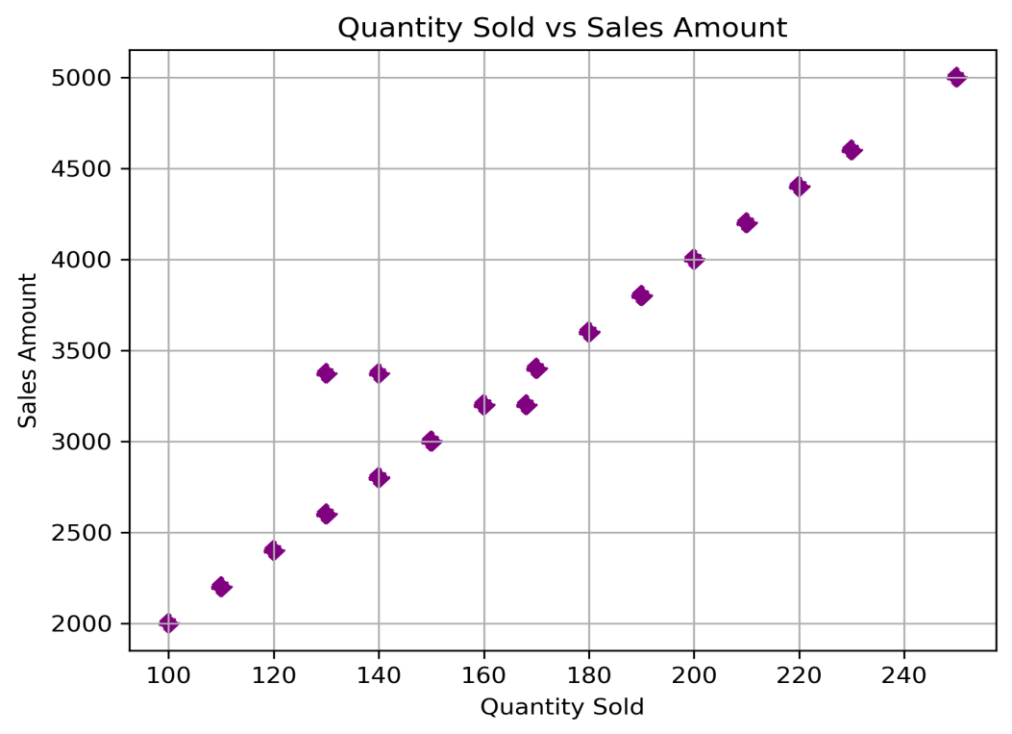
plt.title("Sales Contribution by Region")
plt.savefig("region.png", dpi=300, bbox_inches="tight")
plt.show()
```



# Quantity Sold vs Sales Amount Analysis

```
plt.scatter(df["Quantity Sold"], df["Sales Amount"],color="purple",marker="D",linestyle="-.")

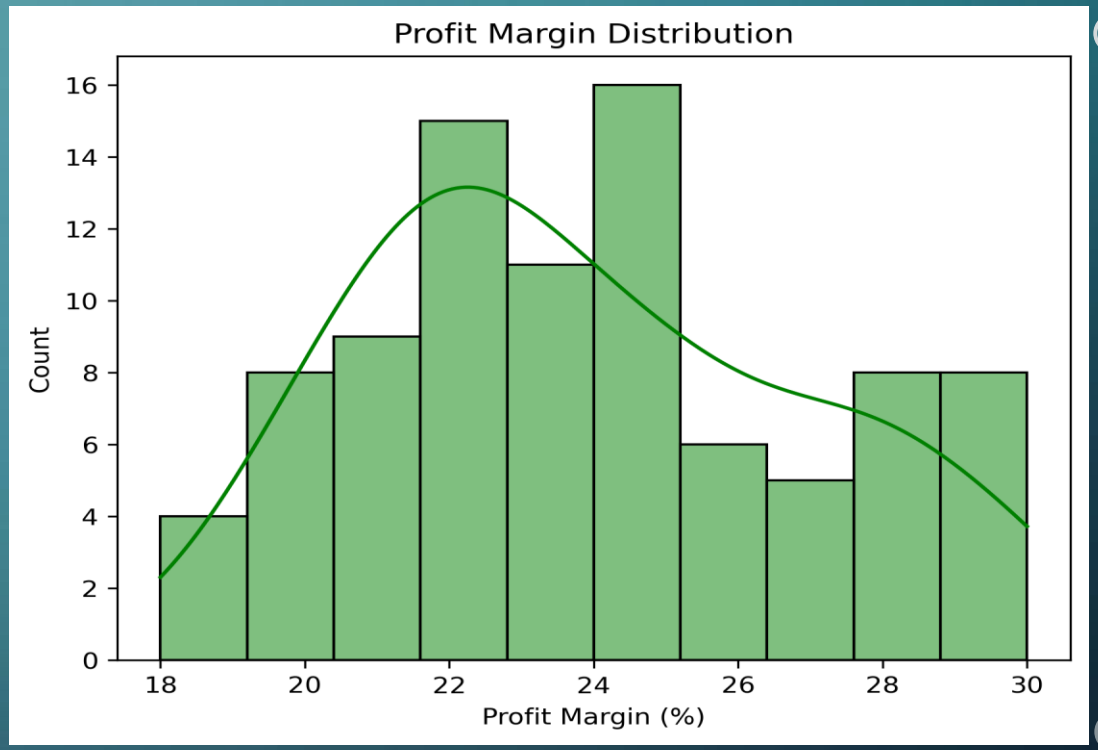

plt.title("Quantity Sold vs Sales Amount")
plt.xlabel("Quantity Sold")
plt.ylabel("Sales Amount")
plt.grid()
plt.savefig("quantity_vs_sales.png", dpi=300, bbox_inches="tight")
plt.show()
```



# Profit Margin Distribution Analysis

```
sns.histplot(
    data=df,
    x="Profit Margin (%)",
    bins=10,
    kde=True,color="green"
)

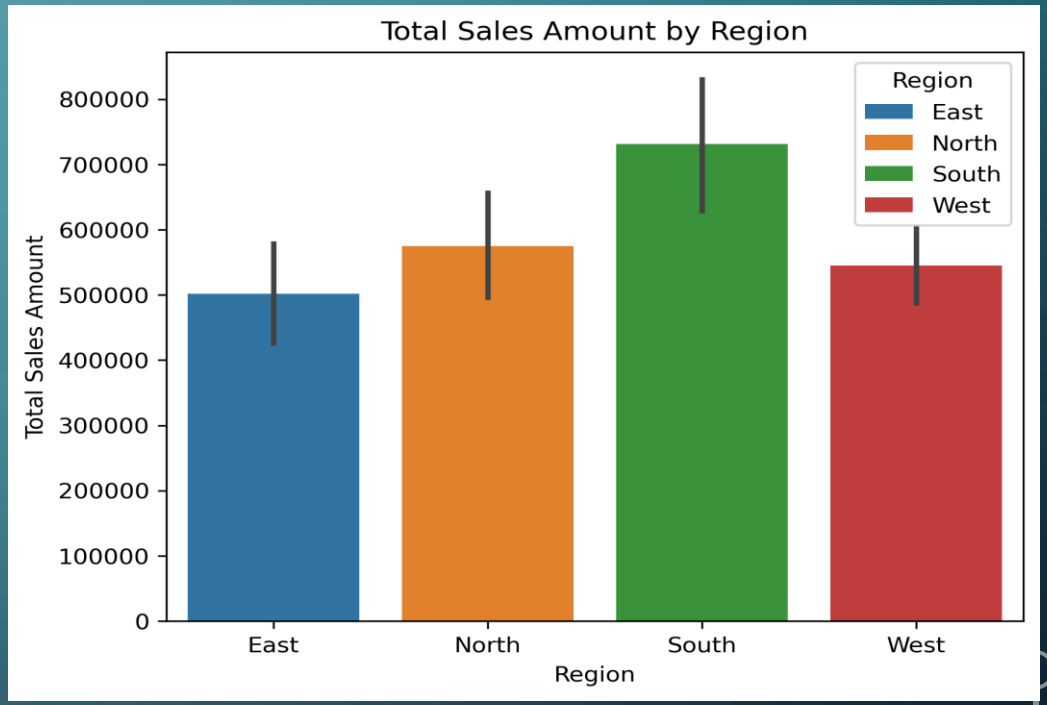
plt.title("Profit Margin Distribution")
plt.savefig("profit.png", dpi=300, bbox_inches="tight")
plt.show()
```



# Total Sales Amount by Region

```
sns.barplot(data=df, x="Region", y="Total Sales Amount",hue="Region",legend=True )


plt.title("Total Sales Amount by Region")
plt.xlabel("Region")
plt.ylabel("Total Sales Amount")
plt.savefig("total sales.png", dpi=300, bbox_inches="tight")
plt.show()
```



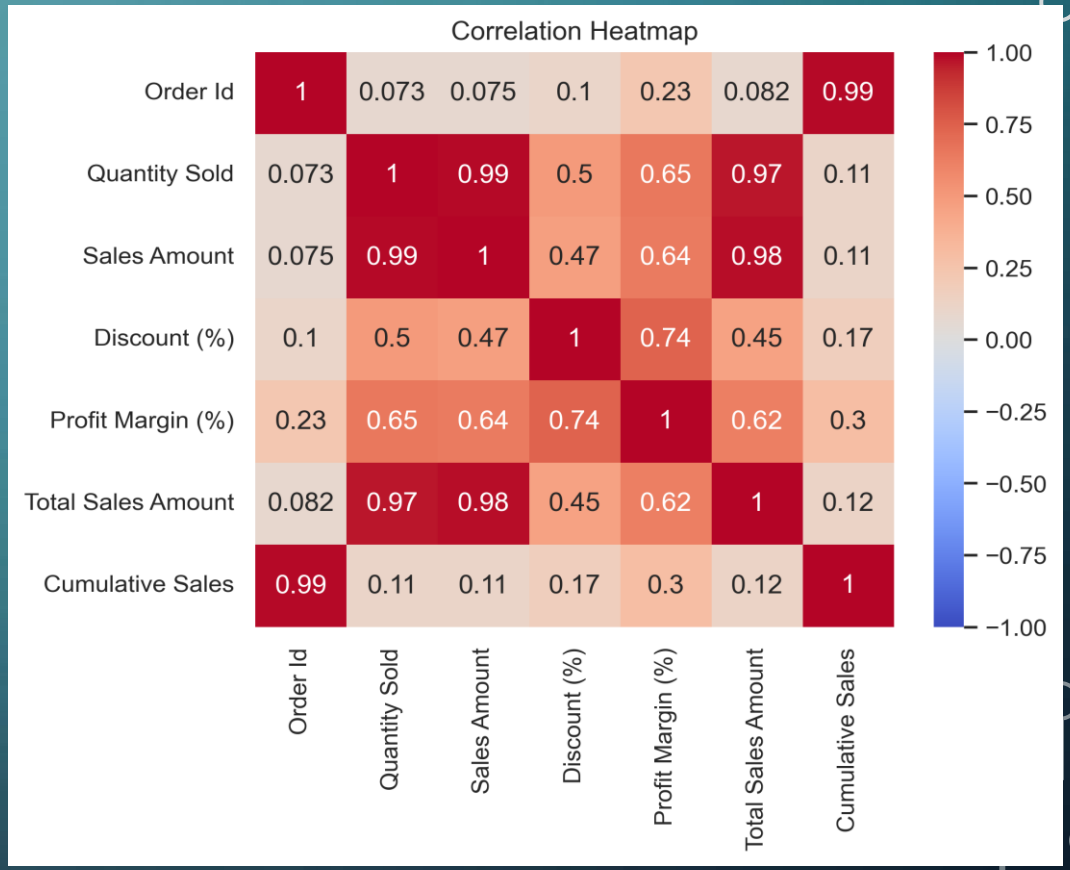
# Correlation Analysis Using a Heatmap

```
numeric_df = df.select_dtypes(include="number")

sns.heatmap(numeric_df.corr(),annot=True,cmap="coolwarm",vmin=-1,vmax=1)

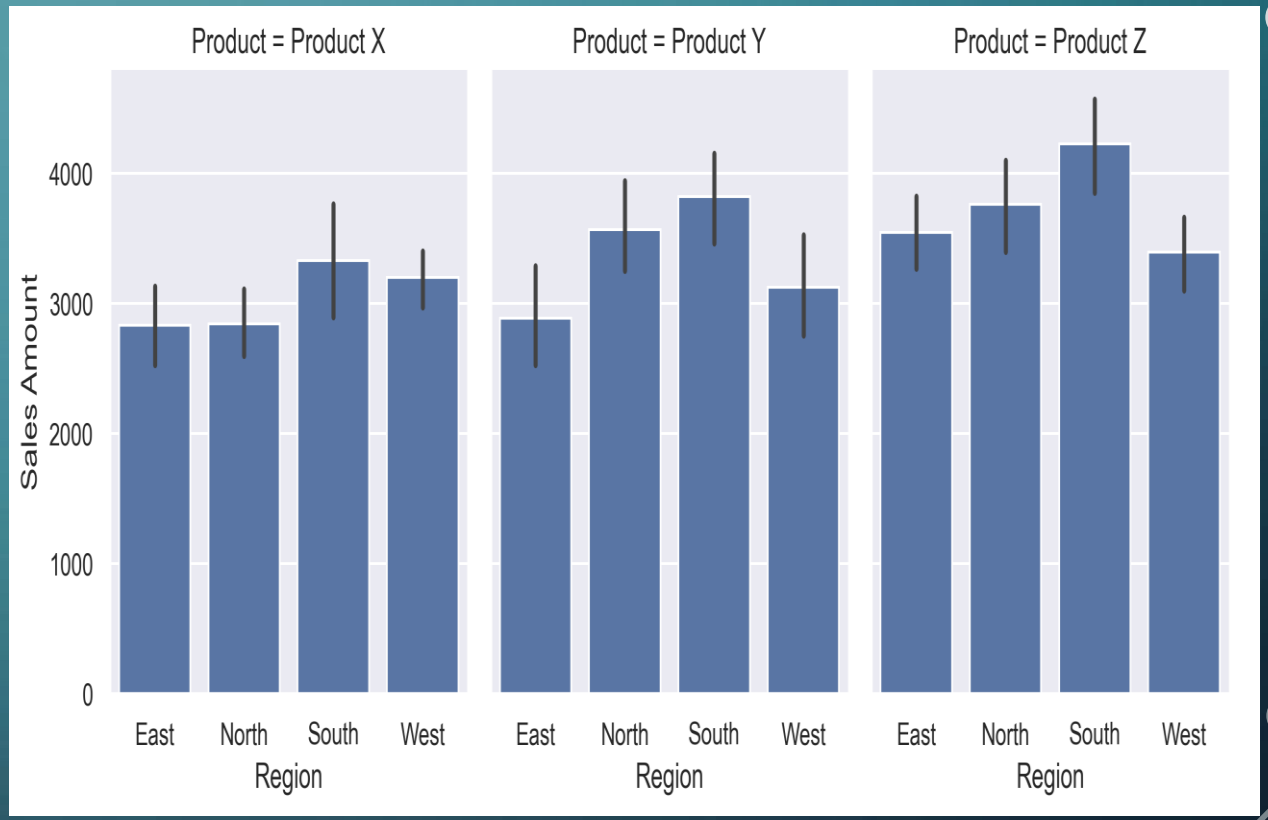
plt.title("Correlation Heatmap")

plt.savefig("heatmap.png", dpi=300, bbox_inches="tight")
plt.show()
```



# Product-wise Sales Performance Across Regions

```
sns.catplot(
    data=df,
    x="Region",
    y="Sales Amount",
    col="Product",
    kind="bar",height=4,aspect=0.8
)
plt.savefig("catplot.png", dpi=300, bbox_inches="tight")
plt.show()
```



# THANK YOU